Gizmo (.GIZ)

Overall

/
- I **Magic** - Always should have a value of 1 (0x01 00 00 00).
- [⧉] **Sections** - Section types listed below.
  - S32 **Section Name**
  - I **Section Length** - Length of the section in bytes.
- I **End of File** - Always should have a value of 0 (0x00 00 00 00)

Special Objects

Special objects are used in multiple gizmo sections and are loaded through the same function. Linking to special objects or other parts can be done without the special objects loading function, but the function is usually used when linking to multiple special objects. Further details on special objects can be found in the .GSC file documentation.

Throughout the rest of the gizmo section documentation, the special object loading *will* be written out completely for each section despite the repetition.

/
- b **Special Object Version**
- b **Special Object Count**
- [⧉] **Special Objects**
  - S8 **Special Object Name**
  - F **Unknown**
  - F **Animation Time**
  - I **Unknown** (Special Object Version >= 2)
  - ? **Additional Loading** - When loading special objects, code for loading additional data can be passed and thus for some special objects loading there may be an extra property.

GizObstacle Section

92 unknown bytes.

GizObstacles are primarily invisible triggers of sorts. For example, in 1-2 when you jump on top of the stone pillars, you enter a GizObstacle trigger that causes the pillar to fall.

/
- b **Version** - This is always at least 10 in Vanilla.

- s **GizObstacle Count**
- [⧉] **GizObstacles**
  - S16 **Name**
  - V3 **Position**
  - V3 **Unknown** (Version >= 2)
  - F **Unknown**
  - F **Unknown**
  - V3 **Unknown** (Version >= 3)
  - s **Unknown** (Version >= 3)
  - I **Unknown**
  - I **Unknown** (Version >= 12)
  - s **Unknown** (Version == 6)
  - b **Unknown** (Version == 6)
  - b **Unknown**
  - b **Unknown**
  - b **Unknown** (Version >= 7)
  - b **Special Object Version**
  - b **Special Object Count**
  - [⧉] **Special Objects**
    - S8 **Special Object Name**
    - F **Unknown**
    - F **Animation Time**
    - I **Unknown** (Special Object Version >= 2)
    - s **Unknown** (Version >= 8)
  - F **Unknown** (Version >= 4)
  - F **Unknown** (Version >= 5)
  - F **Unknown** (Version >= 8)
  - s **Unknown** (Version == 9)
  - S8 **Unknown** (Version >= 10)
  - s **Unknown** (Version >= 9)
  - s **Unknown** (Version >= 9)
  - s **Unknown** (Version >= 9)
  - V3 **Unknown** (Version >= 9)
  - F **Unknown** (Version >= 11)
  - S8 **Unknown** (Version >= 13)
  - S8 **Unknown** (Version >= 14)

GizBuildit Section

16 unknown bytes.

GizBuildit is the buildable attribute applied to special objects.

/

- b **Version**
- s **Buildit Count**
- [⧉] **Buildits**
  - S16 **Name**
  - V3 **Position**
  - b **Special Object Version**
  - b **Special Object Count**
  - [⧉] **Special Objects**
    - S8 **Special Object Name**
    - F **Unknown 1**
    - F **Animation Time**
    - I **Unknown 2** (Special Object Version >= 2)
  - F **Jump Intensity**
  - F **Unknown 1** (Version <= 6)
  - us **Minimum Studs Value?**
  - us **Maximum Studs Value? (Or random variance?)**
  - b **Unknown 2**
  - b **Unknown 3**
  - F **Unknown 4** (Version >= 6)
  - s **Unknown 5** (Version == 7)
  - S8 **Unknown 6** (Version >= 8)
  - s **Studs Pitch** (Version >= 7)
  - s **Studs Yaw** (Version >= 7)
  - V3 **Studs Spawn Position** (Version >= 7)
  - F **Studs Speed** (Version >= 9)
  - s **Unknown 7** (Version >= 4)
  - s **Unknown 8** (Version >= 5)
  - S8 **Unknown 9** (Version >= 5)

GizForce Section

19 unknown bytes.

GizForce is the forceable attribute applied to special objects.

/

- b **Version** - This is always at least 8 in Vanilla, but mostly 16.
- s **Force Count**
- [⧉] **Forces**
  - S16 **Name**
  - V3 **Position**
  - V3 **Unknown** (Version == 1)

- F **Reset Time**
- F **Shake Time** (Version >= 8)
- F **Range**
- V3 **Unknown** (Version == 1)
- s **Unknown** (Version == 1)
- I **Interaction Options** - Needs some further research/experimentation.
  - **Dark Side**
  - **Reset**
- B **Togglable**
- b **Unknown** (Version >= 11)
- b **Unknown**
- b **Unknown** (Version == 1)
- b **Special Object Version**
- b **Special Object Count**
- [⧉] **Special Objects**
  - S8 **Special Object Name**
  - F **Unknown**
  - F **Animation Time**
  - I **Unknown** (Special Object Version >= 2)
  - s **Unknown** (Version >= 9)
- F **Force Speed**
- F **Reset Speed**
- F **Auto Force?** (Version >= 6)
- F **Effect Scale** (Version >= 7)
- F **Unknown** (Version >= 3) - Related to animation?
- s **Unknown** (Version == 4)
- S8 **blowup** (Version >= 5)
- us **Minimum Studs Value?** (Version >= 4)
- us **Maximum Studs Value?** (Version >= 4)
- s **Studs Angle** (Version >= 4)
- V3 **Studs Position** (Version >= 4)
- F **Studs Speed** (Version >= 10)
- S8 **Process Sound** (Version >= 15)
- S8 **Complete Sound** (Version >= 15)
- S8 **Reset Sound** (Version >= 15)

blowup Section

186 unknown bytes.

//test VersionCounter for all unknown values
//attempt to locate stud related values using previous docs

/

- I **Version** - This is always at least 21 in Vanilla.
- I **blowup (Effect) Count** (Version >= 2)
- I **blowup (Property) Count**
- [⧉] **blowups (Effects)** (Version >= 2)
  - S8 **Special Object?**
  - S8 **Name?**
  - S8 **.par Reference?** (Version >= 17)
  - S8 **.par Reference?** (Version >= 17)
  - S8 **.ptl Reference?** (Version >= 4)
  - S8 **.ptl Reference?** (Version >= 4)
  - S8 **.ptl Reference?** (Version >= 4)
  - S8 **_ Reference** (Version >= 26)
  - S8 **_ Reference** (Version >= 26)
  - S8 **_ Reference** (Version >= 27)
  - S8 **_ Reference** (Version >= 27)
  - I **Unknown**
  - I **Unknown** (Version >= 7) - Potentially stud related.
  - b **Unknown** (Version >= 7)
  - F **Unknown** (Version >= 8)
  - S8 **Blowup Decal** (Version >= 9)
  - F **Unknown** (Version >= 14)
  - F **Unknown** (Version >= 14)
  - b **Unknown** (Version >= 15)
  - b **Unknown** (Version >= 15)
  - B **Next Data?** (Version >= 16)
  - V3 **Unknown** (Next Data != 0)
  - F **Unknown** (Next Data != 0)
  - F **Unknown** (Next Data != 0)
  - F **Unknown** (Next Data != 0)
  - F **Unknown** (Next Data != 0)
  - F **Unknown** (Next Data != 0)
  - s **Unknown** (Next Data != 0)
  - b **Unknown** (Next Data != 0)
  - b **Unknown** (Next Data != 0)
  - S8 **Blouwp Emit Object** (Version >= 18)
  - S8 **Blouwp Emit Object** (Version >= 22)
  - S8 **Blouwp Emit Object** (Version >= 22)
  - S8 **Blouwp Emit Object** (Version >= 22)
  - b **Unknown** (Version >= 18)
  - F **Unknown** (Version >= 18)
  - F **Unknown** (Version >= 18)
  - S8 **Blowup Shadow** (Version >= 19)
  - S8 **Blowup Swap** (Version >= 20)

- F **Unknown** (Version >= 23)
- F **Unknown** (Version >= 24)
- [⧉] **blowups (Properties)**
  - S8 **blowup (Effect)?**
  - S8 **Name?** (Version >= 2)
  - V3 **Position**
  - s **Unknown**
  - s **Unknown**
  - s **Unknown**
  - s **Unknown 1a** (Version >= 2 && Version <= 19)
  - I **Unknown 1a** (Version >= 20)
  - I **Unknown 1b** (Version == 28)
  - I **Unknown** (Version >= 30)
  - I **Unknown** (Version >= 2)
  - b **Unknown** (Version >= 2)
  - b **Unknown** (Version >= 2)
  - b **Unknown** (Version >= 4)
  - F **Unknown** (Version >= 6)
  - F **Unknown** (Version >= 8)
  - F **Unknown** (Version >= 8)
  - s **Unknown** (Version >= 9)
  - s **Unknown** (Version >= 9)
  - s **Unknown** (Version >= 9)
  - F **Unknown** (Version >= 9)
  - F **Unknown** (Version >= 9)
  - F **Unknown** (Version >= 9)
  - F **Unknown** (Version >= 10)
  - F **Unknown** (Version >= 11)
  - F **Unknown** (Version >= 11)
  - F **Unknown** (Version >= 11)
  - b **Unknown** (Version >= 12)
  - s **Unknown** (Version >= 13)
  - s **Unknown** (Version >= 13)
  - s **Unknown** (Version >= 19)
  - s **Unknown** (Version >= 19)
  - s **Unknown** (Version >= 19)
  - F **Unknown** (Version >= 19)
  - F **Unknown** (Version >= 19)
  - F **Unknown** (Version >= 19)
  - F **Unknown** (Version >= 19)
  - F **Unknown** (Version >= 21)
  - F **Unknown** (Version >= 23)
  - F **Unknown** (Version >= 31)

GizmoPickup Section

/

- I **Version**
- I **Pickup Count** - The length of the below Pickups array.
- I **Pickup Version?** (Version >= 3) - This value is not entirely understood.
- F **Draw Distance** (Version >= 5) - The distance from the camera pickups must be in to be loaded.
- F **Scale** (Version >= 5) - The scale of the pickups. Although in each level file, this property applies and is consistent through the whole area.
- [⧉] **Pickups**
  - S8 **Pickup Name**
  - V3 **Pickup Position**
  - ≡ **Pickup Type**
    - **default,s - Silver Stud**
    - **g - Gold Stud**
    - **b - Blue Stud**
    - **p - Purple Stud**
    - **m - Minikit**
    - **u - Powerup**
    - **h - Heart**
    - **r - Red Brick**
    - **c - Challenge Minikit**
    - **t - Torpedo**
  - ≡ **Spawn Type** (Version >= 2) - This value has not been entirely researched/experimented with.
    - **default,0 - Normal**
    - **2 - Triggered**
    - **6 - Auto-Collect**
  - b **Spawn Group** (Version >= 4) - Pickups of the same spawn group will all spawn when one is triggered. This is never utilized in Vanilla.

Lever Section

/

- I **Version** - This is always 6 in Vanilla.
- I **Lever Count**
- [⧉] **Levers**
  - S16 **Name**
  - V3 **Position**
  - s **Angle**
  - ≡ **Color {enum?}**
  - B **Multiple Pulls** (Version >= 2)
  - F **Pull Time** (Version >= 3)

- B **Invisible** (Version >= 4)
- V3 **Target Position** (Version >= 5)
- F **Target Size** (Version >= 5)
- B **Target Invisible** (Version >= 6)

Spinner Section

50 unknown bytes.

Spinner is the green and red gizmo that you push to spin.

I

- I **Version** - This is essentially always 9 in Vanilla.
- I **Spinner Count**
- [⧉] **Spinners**
  - S8 **Name**
  - V3 **Position**
  - s **Angle**
  - S8 **Spinner Special Object**
  - b **Unknown Count?**
  - b **Flap Count** (Version >= 2)
  - I **Unknown** (Flap Count != 0 && Version >= 3)
  - F **Unknown** (Flap Count != 0 && Version >= 3)
  - F **Unknown 1** (Flap Count != 0 && Version >= 4)
  - b **Unknown** (Unknown 1 != 0 && Version >= 6)
  - ***Format for Version <= 4 excluded here**
  - b **Special Object Version**
  - b **Special Object Count**
  - [⧉] **Special Objects**
    - S8 **Special Object Name**
    - F **Unknown 1**
    - F **Animation Time**
    - I **Unknown 2** (Special Object Version >= 2)
  - ***Format for Version <= 6 excluded here**
  - [F] **Unknown** (Version >= 7) - Length of Unknown Count
  - F **Unknown** (Version >= 8)
  - F **Unknown** (Version >= 8)

MiniCut Section

50 unknown bytes.

MiniCuts are short cutscenes that occur in the middle of levels.

//Potential values: Trigger bounds, spawn area (after), pause time, tween time, target cam pos, target cam angle?

/

- I **Version** - This is always 1 in Vanilla. Changing this value in TCS does nothing.
- I **MiniCut Count**
- [⧉] **MiniCuts**
  - S8 **Name**
  - F **Unknown**
  - F **Unknown**
  - F **Unknown**
  - F **Unknown**
  - F **Unknown**
  - b **MiniCuts Parts Count**
  - [⧉] **MiniCuts Parts**
    - S8 **MiniCuts Part Name**
    - F **Unknown**
    - F **Unknown**
    - F **Unknown**
    - F **Unknown**
    - s **Unknown**
    - s **Unknown**
    - s **Unknown**
    - F **Unknown**
    - F **Unknown**

Tube Section

Tubes are the areas that blow/fan/float your character upwards. They are also used in the podsprint level.

/

- I **Version** - This is always 2 in Vanilla.
- I **Tube Count**
- [⧉] **Tubes**
  - 🔒 S16 **Name**
  - V3 **Position** - This is at the bottom of the tube.
  - F **Height**
  - F **Radius**
  - B **Magnetic** (Version >= 2)
  - S8 **Special Object** (Version >= 3) - This is not present in TCS.

ZipUp Section

6 unknown bytes.

ZipUps are the red targets and grapples that blaster characters can swing and zip between.

//tests
//analyse for diffs in enemy spawning
//test visibility on two way (town swing)
//script analyse all unknown values

/

- I **Version** - This is always at least 2 in Vanilla, but mostly 4.
- I **ZipUp Count**
- [⧉] **ZipUps**
  - S16 **Name**
  - V3 **Start**
  - V3 **Axis**
  - V3 **End**
  - s **Unknown 1**
  - s **Unknown 2**
  - B **Swing**
  - B **Unknown 3**
  - B **Two Way**
  - B **Invisible** (Version >= 2)
  - B **Unknown 4** (Version >= 3)
  - B **Target(s) Invisible?** (Version >= 4)

GizTurret Section

GizTurrets are the turrets that don't move but shoot at the player, such as in vehicle levels or on Kamino.

/

- b **Version**
- s **GizTurret Count**
- [⧉] **GizTurrets**
  - S16 **Name**
  - b **Special Object Version**
  - b **Special Object Count**
  - [⧉] **Special Objects**
    - S8 **Special Object Name**
    - F **Unknown**

```
│       ├── F Animation Time
│       ├── I Unknown (Special Object Version >= 2)
│       └── s Unknown (Version >= 3) - The TCS loading code for this is a bit odd.
├── V3 Unknown
├── V3 Unknown
├── V3 Unknown
├── V3 Unknown
├── I Unknown
├── I Unknown
├── I Unknown
├── I Unknown
├── I Unknown
├── I Unknown
├── I Unknown (Version >= 2)
├── b Unknown Count
├── [V3] Unknown - Length of Unknown Count.
├── F Unknown
├── F Unknown
├── F Unknown - This is always 0 in Vanilla.
├── F Unknown
├── F Unknown
├── F Unknown
├── s Unknown
├── s Unknown
├── s Unknown
├── V3 Unknown
├── F Unknown (Version >= 6)
├── b Unknown
├── b Unknown (Version >= 4)
├── s Unknown (Version >= 4)
├── S8 Blaster Material?
├── S8 Part?
├── S8 Part?
├── S8 Part? (Version >= 7)
├── S8 Blowup?
└── s Unknown
```

BombGenerator Section

10 unknown bytes.

BombGenerators are the generators that spawn bombs for the player in vehicle levels.

/

- b **Version** - This is always 1 in Vanilla.
- s **BombGenerator Count**
- [⧉] **BombGenerators**
  - S16 **Name**
  - V3 **Position**
  - I **Unknown**
  - b **Special Object Version**
  - b **Special Object Count**
  - [⧉] **Special Objects**
    - S8 **Special Object Name**
    - F **Unknown**
    - F **Animation Time?**
    - I **Unknown** (Special Object Version >= 2)

Panel Section

2 unknown bytes.

/

- I **Version** - This is always at least 5 in Vanilla, but mostly 8.
- I **Panel Count**
- [⧉] **Panels**
  - S32 **Name**
  - V3 **Position**
  - s **Angle**
  - ≡ **Type**
    - **default,0 - Astromech Droid**
    - **1 - Protocol Droid**
    - **2 - Bounty Hunter**
    - **3 - Stormtrooper**
  - B **Invisible** (Version >= 3)
  - V3 **Target Offset** (Version >= 4)
  - F **Target Size** (Version >= 4)
  - B **Target Invisible** (Version >= 5)
  - B **Alternative Face** (Version >= 6) - Only applies to Astromech and Protocol Droid types.
  - B **Alternative Body** (Version >= 6) - Only applies to Astromech and Protocol Droid types. The Protocol Droid type uses the same alternative body as the Astromech type, and thus looks incorrect.
  - B **Unknown 2** (Version >= 7)
  - B **Unknown 3** (Version >= 8) - Almost always false, except for panel3 in deathstarrescue_a.

HatMachine Section

HatMachines are the machines that give your character a hat when the lever is pulled.

/
- I **Version** - This is always 5 in Vanilla.
- I **HatMachine Count**
- [⧉] **HatMachines**
  - S32 **Name**
  - V3 **Position**
  - s **Angle**
  - ≡ **Type**
    - **0 - Random**
    - **1 - Leia**
    - **2 - Fedora**
    - **3 - Top Hat**
    - **4 - Baseball Cap**
    - **default,5 - Stormtrooper**
    - **6 - Bounty Hunter**
    - **7 - Droid Panel**
  - ≡ **Lever Color {enum?}** (Version >= 3)
  - V3 **Target Position** (Version >= 4)
  - F **Target Size** (Version >= 4)
  - B **Target Invisible** (Version >= 5)

PushBlocks Section

4 unknown bytes.

PushBlocks are the blocks that can be pushed and slid across the tiled floors.
Some unknown values may be related to allowing the PushBlock to be pushed after being pushed to a target location.

/
- I **Version**
- I **PushBlocks Count**
- [⧉] **PushBlocks**
  - S8 **Name** - This (likely) links to a special object.
  - F **Snap Range**
  - B **Push Location?** - Whether this is where you are meant to push an object to/on.
  - B **Unknown 1** - This is always 0 in Vanilla.
  - B **Lock Z** - Prevents the PushBlock from being pushed on the Z axis.
  - B **Lock X** - Prevents the PushBlock from being pushed on the X axis.

- B **Unknown 2** (Version >= 4)
- B **Unknown 3** (Version >= 4)
- B **Unknown 4** (Version >= 5) - This is always 0 in Vanilla.
- B **No Slipperiness** (Version >= 5) - Prevents the PushBlock from slipping entirely.
- b **PushBlock Link Object Count** (Version >= 3)
- [S8] **PushBlock Link Objects** (Version >= 3) - These may be special objects.

Torp Machine Section

Torp Machines are the machines that you collect torpedoes from by flying over in vehicle levels.

/

- I **Version** - This is essentially always 3 in Vanilla.
- I **Torp Machine Count**
- F **Scale** (Version >= 3)
- [⧉] **Torp Machines**
  - S32 **Name**
  - V3 **Position**
  - s **Angle**
  - b **Red Outline** (Version >= 2) - Normal black color when 0, red color otherwise.

ShadowEditor Section

52 unknown bytes.

Unknown.

/

- b **Version**
- b **ShadowEdit Count**
- [⧉] **ShadowEdits**
  - V3 **Position?**
  - F **Unknown**
  - F **Unknown** (Version >= 2)
  - F **Unknown** (Version >= 2)
  - F **Unknown** (Version >= 3)
  - F **Unknown** (Version >= 3)
  - F **Unknown** (Version >= 4)
  - F **Unknown** (Version >= 5) - This is always 0 in Vanilla.
  - F **Unknown** (Version >= 5) - This is always 0 in Vanilla.
  - F **Unknown** (Version >= 6)
  - F **Unknown** (Version >= 6)

├── F **Unknown** (Version >= 6)
├── F **Unknown** (Version >= 7)
└── I **Unknown** (Version >= 8) - This is always 0, 1, or 2 in Vanilla.

Grapple Section (Unused. From LB1)

Plug Section (Unused. From LB1)

Techno Section (Unused. From LB1)